# Manage SnapMirror relationship tranfers

ONTAP 9.10.1 REST API Documentation

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# Table of Contents

Manage SnapMirror relationship tranfers . . . 1
SnapMirror relationships relationship.uuid transfers endpoint overview . . . 1
Retrieve ongoing SnapMirror transfers for a relationship . . . 1
Start a SnapMirror transfer operation . . . 12
Retrieve attributes of an ongoing SnapMirror transfer . . . 23
Cancel an ongoing SnapMirror transfer . . . 31

# Manage SnapMirror relationship tranfers

## SnapMirror relationships relationship.uuid transfers endpoint overview

### Overview

This API is used to manage transfers on an existing SnapMirror relationship.

You can initiate SnapMirror operations such as "initialize", "update", "restore-transfer", and "abort" using this API on asynchronous SnapMirror relationship. On a synchronous SnapMirror relationship, you can initiate SnapMirror "initialize" operation. The GET for this API reports the status of both active transfers and transfers that have terminated within the past 24 hours. For the restore relationships, the POST on transfers API triggers "restore-transfer". Successful completion of "restore" also deletes the restore relationship. If the "restore" fails, DELETE on relationships must be called to delete the restore relationship.

## Retrieve ongoing SnapMirror transfers for a relationship

GET `/snapmirror/relationships/{relationship.uuid}/transfers`

**Introduced In:** 9.6

Retrieves the list of ongoing SnapMirror transfers for the specified relationship.

### Related ONTAP commands

- `snapmirror show`

### Example

```
GET "/api/snapmirror/relationships/293baa53-e63d-11e8-bff1-
005056a793dd/transfers"
```

### Learn more

- DOC /snapmirror/relationships/{relationship.uuid}/transfers

### Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| relationship.uuid | string | path | True | Relationship UUID |
| uuid | string | query | False | Filter by uuid |
| checkpoint_size | integer | query | False | Filter by checkpoint_size |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| state | string | query | False | Filter by state |
| snapshot | string | query | False | Filter by snapshot |
| end_time | string | query | False | Filter by end_time<br>• Introduced in: 9.10 |
| total_duration | string | query | False | Filter by total_duration<br>• Introduced in: 9.10 |
| throttle | integer | query | False | Filter by throttle<br>• Introduced in: 9.9 |
| bytes_transferred | integer | query | False | Filter by bytes_transferred |
| error_info.code | integer | query | False | Filter by error_info.code<br>• Introduced in: 9.10 |
| error_info.message | string | query | False | Filter by error_info.message<br>• Introduced in: 9.10 |
| relationship.destination.cluster.name | string | query | False | Filter by relationship.destination.cluster.name<br>• Introduced in: 9.7 |
| relationship.destination.cluster.uuid | string | query | False | Filter by relationship.destination.cluster.uuid<br>• Introduced in: 9.7 |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| relationship.destination.path | string | query | False | Filter by relationship.destination.path |
| relationship.destination.svm.uuid | string | query | False | Filter by relationship.destination.svm.uuid |
| relationship.destination.svm.name | string | query | False | Filter by relationship.destination.svm.name |
| relationship.destination.consistency_group_volumes.uuid | string | query | False | Filter by relationship.destination.consistency_group_volumes.uuid<br>• Introduced in: 9.8 |
| relationship.destination.consistency_group_volumes.name | string | query | False | Filter by relationship.destination.consistency_group_volumes.name<br>• Introduced in: 9.8 |
| relationship.restore | boolean | query | False | Filter by relationship.restore |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of records |
| records | array[snapmirror_transfer] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "bytes_transferred": 0,
    "checkpoint_size": 0,
    "end_time": "2020-12-02T18:36:19-08:00",
    "error_info": {
      "code": 6620046,
      "message": "Transfer aborted"
    },
    "files": {
      "destination_path": "/dirb/file2",
      "source_path": "/dira/file1"
    },
    "relationship": {
      "destination": {
        "cluster": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "cluster1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        },
        "consistency_group_volumes": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "volume1",
```

```
          "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
        },
        "ipspace": "Default",
        "path": "svm1:volume1",
        "svm": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "svm1",
          "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
        }
      },
      "uuid": "d2d7ceea-ab52-11e8-855e-00505682a4c7"
    },
    "snapshot": "string",
    "state": "aborted",
    "throttle": 0,
    "total_duration": "PT28M41S",
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

error_info

Error information for the transfer.

| Name | Type | Description |
|---|---|---|
| code | integer | Error code |
| message | string | Error message |

files

Specifies a file or LUN consisting of a source_path and an optional destination_path. If not specified, the destination_path is the same as the source_path. File restore is not supported if the source_path or destination_path contains commas in its directory or file name.

| Name | Type | Description |
|---|---|---|
| destination_path | string | |
| source_path | string | |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

consistency_group_volumes

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the volume. |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.8 |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

snapmirror_endpoint

Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| consistency_group_volumes | array[consistency_group_volumes] | Mandatory property for a Consistency Group endpoint. Specifies the list of FlexVol volumes for a Consistency Group. |
| ipspace | string | Optional property to specify the IPSpace of the SVM. |

| Name | Type | Description |
| --- | --- | --- |
| path | string | ONTAP FlexVol/FlexGroup - svm1:volume1 ONTAP SVM - svm1: ONTAP Consistency Group - svm1:/cg/cg_name ONTAP S3 - svm1:/bucket/bucket1 NON-ONTAP - objstore1:/objstore<br>• example: svm1:volume1<br>• Introduced in: 9.6 |
| svm | svm | |

relationship

| Name | Type | Description |
| --- | --- | --- |
| destination | snapmirror_endpoint | Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.<br>• Introduced in: 9.6 |
| restore | boolean | Is the relationship for restore? |
| uuid | string | |

snapmirror_transfer

SnapMirror transfer information

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |

| Name | Type | Description |
|---|---|---|
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Start a SnapMirror transfer operation

POST `/snapmirror/relationships/{relationship.uuid}/transfers`

**Introduced In:** 9.6

Starts a SnapMirror transfer operation. This API initiates a restore operation if the SnapMirror relationship is of type "restore". Otherwise, it intiates a SnapMirror "initialize" operation or "update" operation based on the current SnapMirror state.

## Default property values

- `storage_efficiency_enabled` - *true*

## Related ONTAP commands

- `snapmirror update`
- `snapmirror initialize`
- `snapmirror restore`

## Examples

The following examples show how to perform SnapMirror "initialize", "update", and "restore" operations.

Perform SnapMirror initialize or update

```
 POST "/api/snapmirror/relationships/e4e7e130-0279-11e9-b566-
0050568e9909/transfers" '{}'
```

Perform SnapMirror initialize, update or restore with throttle value set

```
 POST "/api/snapmirror/relationships/e4e7e130-0279-11e9-b566-
0050568e9909/transfers" '{"throttle":"100"}'
```

Perform SnapMirror restore transfer of a file

```
 POST "/api/snapmirror/relationships/c8c62a90-0fef-11e9-b09e-
0050568e7067/transfers" '{"source_snapshot": "src",
"files":[{"source_path": "/a1.txt.0", "destination_path": "/a1-
renamed.txt.0"}]}'
```

## Learn more

- DOC /snapmirror/relationships/{relationship.uuid}/transfers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| relationship.uuid | string | path | True | Relationship UUID |
| return_records | boolean | query | False | The default is false. If set to true, the records are returned.<br>• Default value: |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |

| Name | Type | Description |
| --- | --- | --- |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "bytes_transferred": 0,
  "checkpoint_size": 0,
  "end_time": "2020-12-02T18:36:19-08:00",
  "error_info": {
    "code": 6620046,
    "message": "Transfer aborted"
  },
  "files": {
    "destination_path": "/dirb/file2",
    "source_path": "/dira/file1"
  },
  "relationship": {
    "destination": {
      "cluster": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "cluster1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "consistency_group_volumes": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "volume1",
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      },
      "ipspace": "Default",
      "path": "svm1:volume1",
      "svm": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
          },
          "name": "svm1",
          "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
        }
      },
      "uuid": "d2d7ceea-ab52-11e8-855e-00505682a4c7"
    },
    "snapshot": "string",
    "state": "aborted",
    "throttle": 0,
    "total_duration": "PT28M41S",
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  }
```


## Response

```
Status: 201, Created
```

## Error

```
Status: Default
```

ONTAP Error Response codes

| Error code | Description |
|---|---|
| 13303845 | Restore operation failed |
| 13303812 | Initialize operation failed |
| 13303844 | Update operation failed |
| 13303846 | Empty source path file list |
| 13303847 | Invalid arguments |
| 13304040 | Throttle not supported for update of Synchronous SnapMirror relationships |
| 6620237 | SnapMirror relationship database write failed |
| 6620238 | SnapMirror relationship database read failed |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

error_info

Error information for the transfer.

| Name | Type | Description |
|---|---|---|
| code | integer | Error code |
| message | string | Error message |

files

Specifies a file or LUN consisting of a source_path and an optional destination_path. If not specified, the destination_path is the same as the source_path. File restore is not supported if the source_path or destination_path contains commas in its directory or file name.

| Name | Type | Description |
|---|---|---|
| destination_path | string | |
| source_path | string | |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

consistency_group_volumes

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the volume. |

| Name | Type | Description |
| --- | --- | --- |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.8 |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

snapmirror_endpoint

Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| consistency_group_volumes | array[consistency_group_volumes] | Mandatory property for a Consistency Group endpoint. Specifies the list of FlexVol volumes for a Consistency Group. |
| ipspace | string | Optional property to specify the IPSpace of the SVM. |

| Name | Type | Description |
| --- | --- | --- |
| path | string | ONTAP FlexVol/FlexGroup - svm1:volume1 ONTAP SVM - svm1: ONTAP Consistency Group - svm1:/cg/cg_name ONTAP S3 - svm1:/bucket/bucket1 NON-ONTAP - objstore1:/objstore<br>• example: svm1:volume1<br>• Introduced in: 9.6 |
| svm | svm | |

relationship

| Name | Type | Description |
| --- | --- | --- |
| destination | snapmirror_endpoint | Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.<br>• Introduced in: 9.6 |
| restore | boolean | Is the relationship for restore? |
| uuid | string | |

snapmirror_transfer

SnapMirror transfer information

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |

| Name | Type | Description |
|---|---|---|
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Retrieve attributes of an ongoing SnapMirror transfer

GET `/snapmirror/relationships/{relationship.uuid}/transfers/{uuid}`

**Introduced In:** 9.6

Retrieves the attributes of a specific ongoing SnapMirror transfer.

## Related ONTAP commands

- `snapmirror show`

## Example

```
GET "/api/snapmirror/relationships/293baa53-e63d-11e8-bff1-
005056a793dd/transfers/293baa53-e63d-11e8-bff1-005056a793dd"
```

## Learn more

- DOC /snapmirror/relationships/{relationship.uuid}/transfers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| relationship.uuid | string | path | True | Relationship UUID |
| uuid | string | path | True | Transfer UUID |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |

| Name | Type | Description |
| --- | --- | --- |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

**Example response**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "bytes_transferred": 0,
  "checkpoint_size": 0,
  "end_time": "2020-12-02T18:36:19-08:00",
  "error_info": {
    "code": 6620046,
    "message": "Transfer aborted"
  },
  "files": {
    "destination_path": "/dirb/file2",
    "source_path": "/dira/file1"
  },
  "relationship": {
    "destination": {
      "cluster": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "cluster1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "consistency_group_volumes": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "volume1",
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      },
      "ipspace": "Default",
      "path": "svm1:volume1",
      "svm": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
        },
        "name": "svm1",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      }
    },
    "uuid": "d2d7ceea-ab52-11e8-855e-00505682a4c7"
  },
  "snapshot": "string",
  "state": "aborted",
  "throttle": 0,
  "total_duration": "PT28M41S",
  "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

error_info

Error information for the transfer.

| Name | Type | Description |
|---|---|---|
| code | integer | Error code |
| message | string | Error message |

files

Specifies a file or LUN consisting of a source_path and an optional destination_path. If not specified, the destination_path is the same as the source_path. File restore is not supported if the source_path or destination_path contains commas in its directory or file name.

| Name | Type | Description |
|---|---|---|
| destination_path | string | |
| source_path | string | |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

consistency_group_volumes

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the volume. |

| Name | Type | Description |
| --- | --- | --- |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.8 |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

snapmirror_endpoint

Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| consistency_group_volumes | array[consistency_group_volumes] | Mandatory property for a Consistency Group endpoint. Specifies the list of FlexVol volumes for a Consistency Group. |
| ipspace | string | Optional property to specify the IPSpace of the SVM. |

| Name | Type | Description |
| --- | --- | --- |
| path | string | ONTAP FlexVol/FlexGroup - svm1:volume1 ONTAP SVM - svm1: ONTAP Consistency Group - svm1:/cg/cg_name ONTAP S3 - svm1:/bucket/bucket1 NON-ONTAP - objstore1:/objstore<br>• example: svm1:volume1<br>• Introduced in: 9.6 |
| svm | svm | |

relationship

| Name | Type | Description |
| --- | --- | --- |
| destination | snapmirror_endpoint | Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.<br>• Introduced in: 9.6 |
| restore | boolean | Is the relationship for restore? |
| uuid | string | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Cancel an ongoing SnapMirror transfer

PATCH `/snapmirror/relationships/{relationship.uuid}/transfers/{uuid}`

**Introduced In:** 9.6

Aborts an ongoing SnapMirror transfer. This operation is applicable on asynchronous SnapMirror relationships.

## Related ONTAP commands

- `snapmirror abort`

## Example

```
PATCH "/api/snapmirror/relationships/293baa53-e63d-11e8-bff1-
005056a793dd/transfers/293baa53-e63d-11e8-bff1-005056a793dd"
'{"state":"aborted"}'
```

## Learn more

- DOC /snapmirror/relationships/{relationship.uuid}/transfers

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| relationship.uuid | string | path | True | Relationship UUID |
| uuid | string | path | True | Transfer UUID |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |

| Name | Type | Description |
| --- | --- | --- |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "bytes_transferred": 0,
  "checkpoint_size": 0,
  "end_time": "2020-12-02T18:36:19-08:00",
  "error_info": {
    "code": 6620046,
    "message": "Transfer aborted"
  },
  "files": {
    "destination_path": "/dirb/file2",
    "source_path": "/dira/file1"
  },
  "relationship": {
    "destination": {
      "cluster": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "cluster1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "consistency_group_volumes": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "volume1",
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      },
      "ipspace": "Default",
      "path": "svm1:volume1",
      "svm": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
          },
          "name": "svm1",
          "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
        }
      },
      "uuid": "d2d7ceea-ab52-11e8-855e-00505682a4c7"
    },
    "snapshot": "string",
    "state": "aborted",
    "throttle": 0,
    "total_duration": "PT28M41S",
    "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
  }
```


## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response codes

| Error code | Description |
|---|---|
| 13303848 | Abort of sync SnapMirror is not allowed |
| 13303849 | SnapMirror transfer state is invalid |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

error_info

Error information for the transfer.

| Name | Type | Description |
|---|---|---|
| code | integer | Error code |
| message | string | Error message |

files

Specifies a file or LUN consisting of a source_path and an optional destination_path. If not specified, the destination_path is the same as the source_path. File restore is not supported if the source_path or destination_path contains commas in its directory or file name.

| Name | Type | Description |
|---|---|---|
| destination_path | string | |
| source_path | string | |

cluster

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | |
| uuid | string | |

consistency_group_volumes

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the volume. |

| Name | Type | Description |
| --- | --- | --- |
| uuid | string | Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move.<br>• example: 028baa66-41bd-11e9-81d5-00a0986138f7<br>• Introduced in: 9.8 |

svm

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

snapmirror_endpoint

Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.

| Name | Type | Description |
| --- | --- | --- |
| cluster | cluster | |
| consistency_group_volumes | array[consistency_group_volumes] | Mandatory property for a Consistency Group endpoint. Specifies the list of FlexVol volumes for a Consistency Group. |
| ipspace | string | Optional property to specify the IPSpace of the SVM. |

| Name | Type | Description |
| --- | --- | --- |
| path | string | ONTAP FlexVol/FlexGroup - svm1:volume1 ONTAP SVM - svm1: ONTAP Consistency Group - svm1:/cg/cg_name ONTAP S3 - svm1:/bucket/bucket1 NON-ONTAP - objstore1:/objstore<br>• example: svm1:volume1<br>• Introduced in: 9.6 |
| svm | svm | |

relationship

| Name | Type | Description |
| --- | --- | --- |
| destination | snapmirror_endpoint | Endpoint of a SnapMirror relationship. For a GET request, the property "cluster" is populated when the endpoint is on a remote cluster. A POST request to create the destination SVM endpoint or to establish an SVM DR relationship must have the property "cluster" populated with the remote cluster details. A POST request to create the destination FlexVol volume, FlexGroup volume, Consistency Group, ONTAP S3 bucket and NON-ONTAP object-store endpoints can optionally specify the "cluster" property when the source SVM and the destination SVM are peered. A POST request to establish a SnapMirror relationship between the source endpoint and destination endpoint and when the source SVM and the destination SVM are not peered, must specify the "cluster" property for the remote endpoint.<br>• Introduced in: 9.6 |
| restore | boolean | Is the relationship for restore? |
| uuid | string | |

snapmirror_transfer

SnapMirror transfer information

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| bytes_transferred | integer | Bytes transferred |
| checkpoint_size | integer | Amount of data transferred in bytes as recorded in the restart checkpoint. |
| end_time | string | End time of the transfer. |
| error_info | error_info | Error information for the transfer. |
| files | array[files] | This is supported for transfer of restore relationship only. This specifies the list of files or LUNs to be restored. Can contain up to eight files or LUNs. |
| relationship | relationship | |
| snapshot | string | Name of Snapshot copy being transferred. |
| source_snapshot | string | Specifies the Snapshot copy on the source to be transferred to the destination. |
| state | string | Status of the transfer. Set PATCH state to "aborted" to abort the transfer. Set PATCH state to "hard_aborted" to abort the transfer and discard the restart checkpoint. |
| storage_efficiency_enabled | boolean | This is supported for transfer of restore relationship only. Set this property to "false" to turn off storage efficiency for data transferred over the wire and written to the destination. |
| throttle | integer | Throttle in KB/s. Defaults to throttle setting of the associated SnapMirror policy if set, or 0, which is interpreted as unlimited. |

| Name | Type | Description |
| --- | --- | --- |
| total_duration | string | Elapsed transfer time. |
| uuid | string | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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